FreeBSD Jails Orchestration with Ansible

Tutorial with examples

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Contents

1. Introduction
2. FreeBSD collection
3. Examples
4. Install, configure, and activate iocage
5. Plugins iocage
6. Ansible client
7. Infrastructure

Contents (cont.)

8 Appendix

Section 1

Introduction

Why Ansible?

- Preferred for its simplicity due to an agentless model using YAML.
- Used by global leaders (AWS, Cisco, Google, . . .).
- IBM and Red Hat Ansible are closely integrated, particularly through the Red Hat Ansible Automation Platform and the generative AI service, Red Hat Ansible Lightspeed with IBM watsonx Code Assistant.

Does Ansible work with FreeBSD?

Yes. Quoting Ansible documentation BSD efforts and contributions:

```
"BSD support is important to us at Ansible. Even though
the majority of our contributors use and target Linux
we have an active BSD community and strive to be as
BSD-friendly as possible. Please feel free to report
any issues or incompatibilities you discover with BSD;
pull requests with an included fix are also welcome!"
```

See Ansible documentation Managing BSD hosts with Ansible.

Ansible collections

- There are more than 100 collections included in the Ansible distribution.

```
shell> ansible-galaxy collection list | wc -l
107
```

- For example:
  - Amazon.Aws
  - Cisco.Ios
  - Google.Cloud
- See the Collection Index

Ansible collections ansible.*

Tested with FreeBSD:

- ansible.builtin - Modules and plugins contained in ansible-core.
  author: Ansible, Inc
- ansible.posix - For POSIX and POSIX-ish platforms.
  author: Ansible (github.com/ansible)
- ansible.utils - Data management, manipulation, and validation.
  author: Ansible Community

See the release notes what FreeBSD version(s) were tested. For example, quoting v2.20.0:

```
ansible-test - Replace FreeBSD 14.2 with 14.3.
```

Ansible collections community.*

- There are three FreeBSD specific modules in community.general:
  - community.general.sysrc - Manage FreeBSD using sysrc.
  - community.general.pkgng - Package manager for FreeBSD.
  - community.general.portinstall - Installing from FreeBSD's ports system.
- See other collections community.* For example:
  - community.crypto - Modules and plugins for cryptographic operations.
  - community.postgresql - PostgreSQL community modules.
  - community.mysql - MySQL and MariaDB collection.
- These collections are maintained by the Ansible community.

Ansible collection dedicated to FreeBSD is needed to

- support FreeBSD specific subsystems:
  - iocage
  - poudriere
  - bhyve
- support FreeBSD plugins where integration is problematic:
  - service
  - sysctl

Section 2

FreeBSD collection

Proposed FreeBSD collection

- Ansible Galaxy:
  https://galaxy.ansible.com/ui/repo/published/vbotka/freebsd/
- GitHub:
  https://github.com/vbotka/ansible-collection-freebsd/
- Read The Docs:
  https://ansible-collection-freebsd.readthedocs.io/en/latest/

Collection Content

- Plugins:
  - module iocage - iocage jail handling.
  - module service - Control or list system services.
  - module ucl - CRUD-like interface for managing UCL files.
  - inventory iocage - iocage inventory source.
  - filter iocage - Parse iocage lists.
  - lookup galaxy_info - Get the meta data from galaxy.yml
- Roles
- Playbooks

Note: The proposed FreeBSD collection is a work in progress.

We focus on the `iocage` plugins

- inventory iocage - iocage inventory source.
- module iocage - iocage jail handling.
- filter iocage - Parse iocage lists.

inventory iocage

- Get inventory hosts from the `iocage` jail manager running on `host`.
- Properly documented in ansible-collection-freebsd.readthedocs.io
- Included also in the collection community.general as community.general.iocage
- Also documented in docs.ansible.com
- License GPLv3
- Extends fragments:

```
extends_documentation_fragment:
  - ansible.builtin.constructed
  - ansible.builtin.inventory_cache
```

inventory iocage - Basics

- Get the inventory from the command line

```
shell> ssh admin@10.1.0.73 iocage list -l
```

- Configure inventory plugin `hosts/02_iocage.yml`

```yaml
plugin: vbotka.freebsd.iocage
host: 10.1.0.73
user: admin
```

- Display the inventory

```
shell> ansible-inventory -i hosts/02_iocage.yml --list --yaml
```

inventory iocage - DHCP

- As admin at the controller, list the jails. The IP4 tab says "... address requires root". Use sudo if enabled

```
shell> ssh admin@10.1.0.73 sudo iocage list -l
```

- Update the inventory configuration `hosts/02_iocage.yml`. Use the parameter `sudo`

```yaml
plugin: vbotka.freebsd.iocage
host: 10.1.0.73
user: admin
sudo: true
```

- Display the inventory

```
shell> ansible-inventory -i hosts/02_iocage.yml --list --yaml
```

inventory iocage - Hooks

- When `sudo` is not enabled, update the inventory configuration `hosts/02_iocage.yml`. Use the parameter `hooks_results`

```yaml
plugin: vbotka.freebsd.iocage
host: 10.1.0.73
user: admin
hooks_results:
  - /var/db/dhclient-hook.address.epair0b
```

- Example of a hook `/etc/dhclient-exit-hooks`

```
case "$reason" in
    "BOUND"|"REBIND"|"REBOOT"|"RENEW")
    echo $new_ip_address > /var/db/dhclient-hook.address.$interface
    ;;
esac
```

- Display the inventory

```
shell> ansible-inventory -i hosts/02_iocage.yml --list --yaml
```

inventory iocage - Properties

Optionally, get the `iocage properties`. Update the inventory configuration `hosts/02_iocage.yml`. Use the parameter `get_properties`. Compose the variable `ansible_host`

```yaml
plugin: vbotka.freebsd.iocage
host: 10.1.0.73
user: admin
get_properties: true
hooks_results:
  - /var/db/dhclient-hook.address.epair0b
compose:
  ansible_host: (iocage_hooks.0 == '-') |
                ternary(iocage_ip4, iocage_hooks.0)
```

inventory iocage - Tags

Use `property notes` to store `tags`. Update the inventory configuration `hosts/02_iocage.yml`. Compose the dictionary `iocage_tags` and create `groups`. The `properties` are required. Enable the parameter `get_properties`

```yaml
plugin: vbotka.freebsd.iocage
host: 10.1.0.73
user: admin
get_properties: true
hooks_results:
  - /var/db/dhclient-hook.address.epair0b
compose:
  ansible_host: (iocage_hooks.0 == '-') |
                ternary(iocage_ip4, iocage_hooks.0)
  iocage_tags: dict(iocage_properties.notes |
                split | map('split', '='))
keyed_groups:
  - prefix: vmm
    key: iocage_tags.vmm
  - prefix: project
    key: iocage_tags.project
```

inventory iocage - Aliases

The tag `alias` is used to create inventory aliases. Update the inventory configuration `hosts/02_iocage.yml`. Set the parameter `inventory_hostname_tag` to `alias`. This tag keeps the value of the inventory alias.

```yaml
plugin: vbotka.freebsd.iocage
host: 10.1.0.73
user: admin
get_properties: true
inventory_hostname_tag: alias
hooks_results:
  - /var/db/dhclient-hook.address.epair0b
compose:
  ansible_host: (iocage_hooks.0 == '-') |
                ternary(iocage_ip4, iocage_hooks.0)
  iocage_tags: dict(iocage_properties.notes | split | map('split', '='))
keyed_groups:
  - prefix: vmm
    key: iocage_tags.vmm
  - prefix: project
    key: iocage_tags.project
```

module iocage

- FreeBSD iocage jail handling.
- See the in-line documentation on galaxy.ansible.com
- Implements most of the `iocage` options.
- See the in-line examples.

filter iocage

- This filter parses `iocage` list output.
- See the in-line documentation on galaxy.ansible.com
- See the in-line examples.

Section 3

Examples

Example groups

- Install, configure, and activate `iocage`
- Plugins `iocage`
- Ansible client
- Infrastructure

How to use the examples

Each example provides links to the:

- example source code
- detailed description of the example
- example results

To test the examples:

- clone the source code repository
- open the guide with the detailed descriptions
- run the examples and compare the results

Notes

- All examples comprise additional files not shown in the file' tree. See them for more details.
- Most examples comprise `batch.sh` that runs the commands and creates the output.
- Most plays in `batch.sh` are idempotent. The output of such a play may show status `ok` instead of expected `changed` if the play has already been run.
- The playbooks in the examples use dashes - in their filenames. For example, `pb-iocage.yml`.
- The playbooks in the collection, because of the Ansible collection naming conventions, use underscores _ in their filenames. For example, `pb_iocage_template.yml`.

Section 4

Install, configure, and activate iocage

example 001: Install iocage

Use the role `vbotka.freebsd.iocage` to install the package `iocage`.

requirements:

- root privilege in the managed nodes

links:

- source code
- results

example 002: Activate `iocage`

Use the role `vbotka.freebsd.iocage` to activate `iocage`.

requirements:

- root privilege in the managed nodes
- binary iocage

links:

- source code
- results

example 003: Audit `iocage` host

Use the role `vbotka.freebsd.iocage` to audit the `iocage` configuration.

requirements:

- root privilege in the managed nodes
- binary iocage

links:

- source code
- results

Section 5

Plugins iocage

example 010: Clone basejails and create inventory

Fetch releases, create basejails, clone jails from the basejails, and start the jails. Use the inventory plugin `vbotka.freebsd.iocage` to create the inventory. Display the created inventory.

requirements:

- module `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- activated binary `iocage`

links:

- source code
- results

example 011: Display variables `iocage_*`

extends: example 010

Display all variables `iocage_*` created by the inventory plugin `vbotka.freebsd.iocage`.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- jails created in example 010

links:

- source code
- results

example 012: Display `iocage_properties`

extends: example 010

Enable and display `iocage_properties`.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- jails created in example 010

links:

- source code
- results

example 013: Tags and custom groups

extends: example 010

Use the property `notes` to create tags:

- Add the property `notes: "vmm=localhost"`

In the inventory plugin:

- compose the variable `iocage_tags`
- create groups `vmm_*` from the attribute `iocage_tags.vmm`

requirements:

- module `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- activated binary `iocage`
- fetched releases

links:

- source code
- results

example 014: Inventory cache

Enable and test inventory cache.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- jails created in example 010

links:

- source code
- results

example 015: Multiple inventory cache

Enabled cache in multiple inventory files.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- jails created in example 010

links:

- source code
- results

example 016: Multiple inventory constructed

Create inventory groups using the inventory plugin `ansible.builtin.constructed` after the two inventory plugin `vbotka.freebsd.iocage` configuration files.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- jails created in example 010

links:

- source code
- results

example 017: community.general.iocage

Use the inventory plugin `community.general.iocage` instead of the inventory plugin `vbotka.freebsd.iocage`.

links:

- source code
- results

example 018: Clone basejails. Use DHCP.

Use DHCP to configure the interfaces.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- jails created in example 010

links:

- source code
- results

example 019: Inventory option use_vars_plugins

The option `use_vars_plugins`, responsible for reading `host_vars` and `group_vars` directories, is not available in the inventory plugin `vbotka.freebsd.iocage` because the `constructed fragment` doesn't provide it.

- Use the inventory plugin `ansible.builtin.constructed` to read `group_vars`.
- Use the variable `region` to create the groups `region_EU` and `region_US`.

links:

- source code
- results

example 020: Get inventory aliases from notes

Get the `inventory aliases` from the `iocage property notes`. In the inventory plugin `vbotka.freebsd.iocage`, use the option `inventory_hostname_tag` to tell the plugin which tag to use.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- templates created in example 202

links:

- source code
- results

example 030: Create custom facts

extends: example 020

Create custom facts to provide a dictionary of iocage datasets lists. Use the filter `vbotka.freebsd.iocage` to parse them.

requirements:

- role `vbotka.freebsd.iocage`
- filter `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- jails created in previous examples

links:

- source code
- results

Section 6

Ansible client

example 200: Create iocage templates. Clone jails.

- Create iocage templates for Ansible clients.
- Clone jails.

requirements:

- playbook `vbotka.freebsd.pb_iocage_template.yml`
- playbook `vbotka.freebsd.pb_iocage_ansible_clients.yml`
- module `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- activated `iocage`
- fetched releases

links:

- source code
- results

example 201: Display iocage datasets

- Get and display `iocage` datasets.

links:

- source code
- results

example 202: Create iocage templates. Clone DHCP jails

extends: example 200

- Create iocage templates for Ansible clients.
- Get the IP addresses by DHCP.
- Create the `dhclient-exit-hooks`.

requirements:

- playbook `vbotka.freebsd.pb_iocage_template.yml`
- playbook `vbotka.freebsd.pb_iocage_ansible_clients.yml`
- module `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- activated `iocage`
- fetched releases

links:

- source code
- results

example 203: Create DHCP jails with auto UUID and iocage_tags

- Automatically generate the jails UUID names. At each iocage host, create three jails from the template `ansible_client` The module `vbotka.freebsd.iocage` doesn't work with multiple names. Use `ansible.builtin.command` instead.
- In the inventory plugin, compose the variable `iocage_tags` iocage_tags: dict(iocage_properties.notes | split | map('split', '='))
- Create groups from `iocage_tags`

requirements:

- playbook `vbotka.freebsd.pb_iocage_ansible_clients.yml`
- module `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- templates created in example 202

links:

- source code
- results

example 204: Create DHCP jails with auto UUID and iocage_tags v2

extends: example 203

- Instead of the module `vbotka.freebsd.iocage` create the variable `iocage_jails` using the filter `vbotka.freebsd.iocage`

requirements:

- filter `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- templates created in exmaple 202

links:

- source code
- results

example 206: Create DHCP and fixed IP jails

extends: example 203

- In the inventory plugin `vbotka.freebsd.iocage` configuration file, use the option `hooks_results` to get the DHCP IP address.

requirements:

- playbook `vbotka.freebsd.pb_iocage_ansible_clients.yml`
- module `vbotka.freebsd.iocage`
- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- templates created in example 202

links:

- source code
- results

example 207: Create DHCP jails with auto UUID, iocage_tags, alias and class

- At multiple iocage hosts, create and run VNET jails with a DHCP interface from the template `ansible_client`.
- Use the dictionary `iocage_tags` and option `inventory_hostname_tag` to create `inventory aliases`.
- Group the jails by iocage hosts, states, and classes.
- Declare the project in a single dictionary. The dictionary keys are jails' aliases.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- templates created in example 202

links:

- source code
- results

example 208: Create iocage template for ansible-pull

- Create iocage template `ansible_client_pull` that will use `ansible-pull`.

requirements:

- playbook `vbotka.freebsd.pb_iocage_template.yml`
- module `vbotka.freebsd.iocage`
- root privilege in the managed nodes

links:

- source code
- results

example 209: Create iocage pkglist file

- Use the role `vbotka.freebsd.iocage` to create `iocage_` list of packages for `Automatic Package Installation`.
- Create Ansible template for `Apache HTTP server`.

requirements:

- role `vbotka.freebsd.iocage.yml`
- playbook `vbotka.freebsd.pb_iocage_template.yml`

links:

- source code
- results

example 210: Test empty iocage notes

- Test empty iocage notes. Create `iocage_tags`. The result should be an empty dictionary.

requirements:

- inventory plugin `vbotka.freebsd.iocage`

links:

- source code
- results

Section 7

Infrastructure

example 500: syslog-ng server and syslog-ng clients

Configure and run a log server. Configure log clients and test them. Use `syslog-ng`. Use the jails created in the example 207 Create DHCP jails with auto `UUID`, `iocage_tags`, `alias` and `class`. The project keys are jail's aliases.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- module `vbotka.freebsd.service`
- role `vbotka.freebsd.postinstall`
- jails created in example 207

links:

- source code
- results

example 501: iocage host

Configure iocage host.

requirements:

- role `vbotka.freebsd.iocage`
- role `vbotka.freebsd.network`
- role `vbotka.freebsd.pf`
- role `vbotka.freebsd.postinstall`
- role `vbotka.freebsd.zfs`

links:

- source code
- results

example 502: Branch server

Install and configure syslog-ng and git servers in the branch-server.

requirements:

- role `vbotka.freebsd.config_light`
- role `vbotka.freebsd.postinstall`

links:

- source code
- results

example 510: Project ansible-pull

Use the template `ansible_client_pull` to create project jails.

requirements:

- inventory plugin `vbotka.freebsd.iocage`
- root privilege in the managed nodes
- templates `ansible_client_pull` created in example 208

links:

- source code
- results

Section 8

Appendix

App1. FreeBSD collection configuration

- In the repository, see the tree

  ```
  shell> tree setup
  ```

- See the playbook *setup/.configure.yml* . This play can:
  - create setup/vars/roles_latest.yml
- See the file *setup/vars/roles_latest.yml*
- This play should be used to configure other options when necessary.

App2. FreeBSD collection setup

- In the repository, see the tree

```
shell> tree setup
```

- See the playbook *setup/setup.yml*. This play can:
  - display variables
  - install plugins
  - install roles
- See the files in the directory *setup/vars/*

App3. Recommended reading

- FreeBSD Handbook: Chapter 17. Jails and Containers
- FreeBSD Forum: Jails
- FreeBSD Mastery: Jails
- FreeBSD Mastery: ZFS